on Moon and Mars by generating comprehensive geological map, mineral map and quantification of morphological features using multi-sensor planetary datasets to understand their nature of occurrence, associated mineralogy, morphological characteristic and geological evolution.

RES-IIRS-2025-003

Name of ISRO/DOS Centre/Unit

Indian Institute of Remote Sensing, Dehradun

Title of the research proposal

Quantifying Methane Emissions from Indian Wetlands using Satellite Datasets

Area of Research

RS & GIS

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Summary of the Proposed Research

Wetlands are one of the major natural sources of methane (CH), a potent greenhouse gas with high radiative forcing. Quantifying methane emissions from wetlands is important for understanding India's greenhouse gas balance, informing climate policy, and improving models of methane flux under changing land use and climate.

There have been several studies estimating wetland methane emissions in India through field measurements (flux chambers, eddy covariance, etc.) and through broader-scale modelling using coarse satellite data. For example:

- A hierarchical empirical model using MODIS (Moderate Resolution Imaging Spectroradiometer) data and ARIMA modeling for wetland extent, temperature etc.
- Case studies in Odisha (Orissa) for Chilka, Anshupa, and Gahirmatha estimating seasonal and annual CH fluxes.
- Recent inventories of methane emissions from various sources in India indicating wetlands account for ~8.6% of national CH emissions.
- However, these approaches typically suffer from limitations such as:
- Spatial resolution: Many sensors (MODIS, Sentinel-5, etc.) are coarse, making it hard to resolve emissions at fine scales (e.g. patches of wetland, small water channels).
- Spectral resolution: Detecting methane with confidence often requires narrow-band SWIR near methane absorption features (e.g., ~1.65 µm, ~2.3 µm).
- Temporal revisits: Wetlands being dynamic (hydrology, inundation, vegetation) require frequent observations.
- Cloud cover, atmospheric interference: Wet season, monsoons, etc., complicate remote sensing.

Recent advances:

- Hyperspectral satellites like EnMAP, PRISMA, EMIT and the commercial constellation (GHGSAT, PIXXEL Firefly constellation, Planet's Tanager series) have been demonstrated for methane detection and quantification.
- The high spatial and fine spectral resolution (narrow SWIR absorption bands), is capable of detecting point sources with emission rates on the order of ~100 kg CH per hour under good conditions.

Given this, there is a strong opportunity to apply satellite datasets to Indian wetlands, both to improve emission estimates locally, and to compare with other hyperspectral (and multispectral) systems to understand trade offs (e.g., detection limit vs coverage vs revisit rate).

Scope of the work:

- Quantify methane emissions from representative Indian wetlands using remote sensing data, over a multi-year period.
- Compare the results (spatial, temporal, magnitude) with estimates using hyperspectral datasets and possibly multispectral sensors.
- Assess detection limits, uncertainties, and spatial/temporal dynamics of wetland CH fluxes as observed through hyperspectral remote sensing.
- Evaluate scaling: how local emissions inferred via hyperspectral and other sensors aggregate to regional and national scales; comparison with bottom up/field measurement inventories.
- Develop methodological protocols for remote sensing of methane emissions in wetlands: best practices for data acquisition, atmospheric correction, cloud masking, plume detection, etc.

Linkages to Space Programme:

- NICES

Expected Deliverables:

- High resolution quantification of methane emissions from multiple Indian wetlands across seasons.
- Comparative assessment of hyperspectral and other datasets: strengths, weaknesses, detection thresholds, etc.
- Improved national / regional methane emission estimates from wetlands with lower uncertainty.
- Methodological protocols for wetland CH remote sensing (useful for researchers, government, satellite mission planners).
- Publications in peer reviewed journals; datasets made available (subject to licensing) for the community.
- Potential policy relevance: better information for India's climate reporting, potential mitigation strategies for high emission wetlands or hotspots.

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Name of ISRO/DOS Centre/Unit

Indian Institute of Remote Sensing, Dehradun